Sea Turtle Conservation AI Project

Bryce Addeo

AI Fundamentals – EEL3872

Professor Diego Alvarado

25 November 2024

Table of Contents

Abstract……..Pg. 3

Introduction & Problem Statement……Pg. 3

Assumptions……..…Pg. 4

Data Input……..Pg. 4

Model Conceptualization & Presentation……..Pg. 8

Verification & Validation…….....Pg. 12

Experimentation……...Pg. 15

Summary and Recommendations……Pg. 17

References……Pg. 18

Appendix……..Pg. 18

Abstract

The Sea Turtle Conservation AI Project aimed to aid efforts in sea turtle conservation by leveraging artificial intelligence to enhance public knowledge, improve measurement techniques and enable species identification of sea turtles. Through the training of an AI bot named Cammy, the project demonstrated the power of supervised and unsupervised learning to perform three distinct tasks. First, Cammy was trained using supervised learning to answer commonly asked questions about sea turtles, enabling more effective communication and education. Second, a regression model was applied to predict sea turtle weights based on species, carapace length, and carapace width, reducing the need for invasive measurement techniques. Finally, classification and unsupervised learning were used to train Cammy to accurately identify sea turtle species from photographs to make it easier for researchers to analyze and group data. While Cammy performed well overall, the bot still made a few errors suggesting there is room for improvement and further training. Despite the errors, the project undoubtedly proved the potential of AI in advancing wildlife conservation.

Introduction & Problem Statement

A lack of knowledge on sea turtles and improper data collecting methods has put many species of sea turtles in danger. Therefore, I have been tasked to develop an AI bot to aid in sea turtle conservation by answering questions, predicting weights, and identifying species of sea turtles.

The goal of part one of the Sea Turtle Conservation AI Project was to train a chat bot to answer various questions about sea turtles. This was done at CogUniversity which is an online school dedicated to teaching machines rather than humans. For this project, a cog bot named Cammy was selected to be trained to answer questions about sea turtles so that it could ultimately teach humans more about them and show how they can help nurture the population. However, to effectively train Cammy, I needed to input various questions and answers with rankings to demonstrate how to provide proper responses. This is an example of supervised machine learning that is a necessary step in training many AI systems.

The goal of part two of the project was to train the chat bot, Cammy, to provide weight predictions for a sea turtle when given the species, carapace length, and carapace width. This is important because it will allow researchers to obtain critical measurements without the risk of injuring the turtles by measuring them. A stereo camera can determine the length and width of the carapace and then Cammy can determine the weight based off that data. This is an example of regression which is a form of supervised learning where a dependent variable can be predicted (in this case weight) from known independent variables (in this case carapace length and width).

Finally, the goal of part three of the project was to train Cammy to identify the species of a sea turtle when given a random photograph of one. This is important because it will allow

researchers to easily identify the species of sea turtle they are dealing with if they are not already experts in the field. This is an example of both supervised and unsupervised learning. First, supervised learning, specifically classification, is used because the data is externally labeled and then Cammy uses this labeled data to train to be able to label new images. Next, unsupervised learning occurs when the AutoKeras algorithm teaches Cammy how to specifically analyze an image to output a species and probability of accuracy.

Assumptions

Before training Cammy could begin, some key assumptions had to be made. First off, the data provided for use was assumed to be accurate and sufficient for training. For example, the questions were assumed to be the most common ones asked about sea turtles, the measurement and weight data reflected mostly average sea turtle weights but also included outliers, and the species photos showed enough different variations of sea turtle appearances per species. Next, it was assumed that the algorithms uploaded to Cammy were the correct ones to meet the goals of the project. They would later be tested to find out the accuracy of this assumption.

Data Input

The process of training Cammy began with uploading 40 questions through a CSV Microsoft Excel file. These were all common questions about sea turtles shown in Figure 1 below.

	question
2	How many eggs do turtles lay
3	what is the biggest sea turtle ever
4	When do sea turtles lay there eggs
5	Can I watch a sea turtle lay eggs without disturbing them
6	How do the baby turtles know the way to the sea
7	What is the life expectancy of a sea turtle
8	What if see a turtle laying its eggs
9	How deep can a sea turtle go
10	how many eggs do turtles lay at a time
11	What to turtles eat
12	what time of year do sea turtles lay eggs
13	What guides a baby sea turtle to the water
14	How deep can sea turtles dive?
15	What should I do if I sea a sea turtle laying eggs on the beach
16	What is the sea turtles favorite food
17	How big was the largest sea turtle ever recorded?
18	Can I watch the moma sea turtle lay eggs or will that disturb them
19	what is the world record sea turtle
20	What to sea turtles eat
21	Can sea turtles swim down deep
22	How old do sea turtles get
23	What do sea turtles like to eat
24	How long to turtles live
25	When do sea turtles nest?
26	When do turtles nest.
27	How can I watch a sea turtle lay eggs without bothering them
28	How does the freshly hatched sea turtle who which way to walk to ge
29	How many eggs are in a turtle nest
30	How big can a sea turtle be
31	How do sea turtles locate the ocean while they are on the beach?
32	how big do sea turtles get
33	what is their nesting season
34	how many sea turtle eggs do you find in a single nest
35	How long do sea turtles live?
36	How many eggs do sea turtles lay?
37	Do sea turtles live a long time
38	What is the max depth a sea turtle can go
39	How deep can sea turtles swim
40	What do the sea turtles like to eat
41	How do the baby turtles know which way to walk to get to the ocean

Figure 1: Sea Turtle Questions

Although there were 40 questions, only about 8-10 of them were completely unique while the others were just the same questions worded a little differently. Therefore, the same answer could be provided to Cammy for a number of questions, making the training process quicker.

Next, in part two, the process of training Cammy to predict the weight of sea turtles began with uploading a Microsoft Excel CSV file that contained data for sea turtle species, carapace length (cm), carapace width (cm) and weight (kg). Below is a small portion of this data from the uploaded file.

	A	B	C	D
1	Species	Straight_Carapace_Length_cm	Straight_Carapace_Width_cm	weight_kg
2	loggerhead	59.4	48.9	32.4
3	loggerhead	4.41	3.33	17.3
4	loggerhead	61.1	52.4	36.4
5	green	33.4	26.4	4.5
6	green	36.2	30	5.1
7	loggerhead	54.4	44.9	24.4
8	ridley	30.4	28	4
9	loggerhead	72.5	56.7	28.9
10	loggerhead	66.6	54.7	45
11	green	40.5	31.7	7.2
12	loggerhead	64.2	51.3	37.2
13	loggerhead	63.2	50.2	32.8
14	loggerhead	4.88	3.93	24.3
15	ridley	29.9	27.9	3.7

Figure 2: Sample of Sea Turtle Measurement Data

This data file contained measurements for 5,698 different sea turtles and serves the basis of what Cammy will use to develop relationships between the variables to make the best predictions. However, before training Cammy to use this data, it was important for me to better understand the data myself. Therefore, I made the data report shown below on Google Looker Studio.

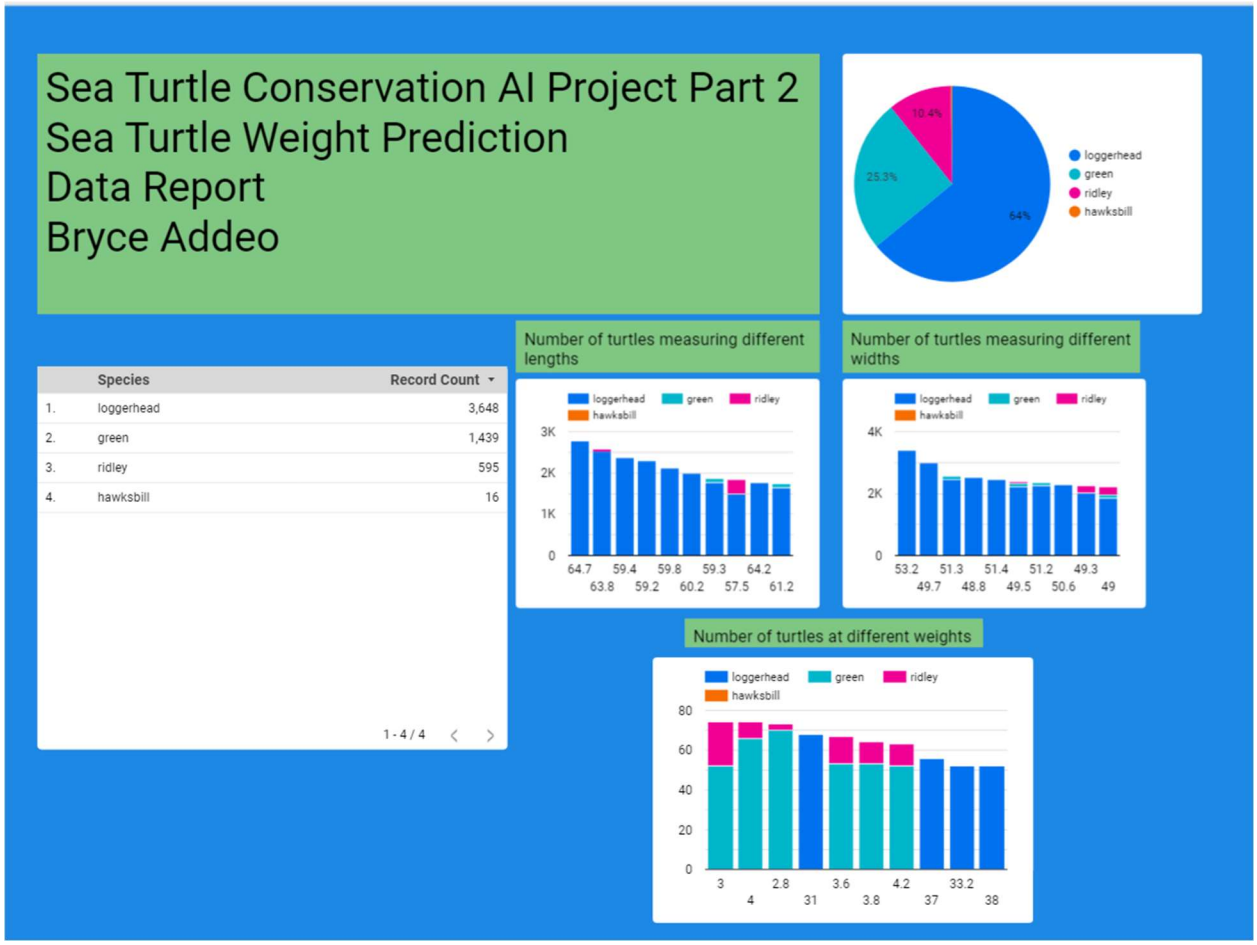

Figure 3: Sea Turtle Measurement Data Report

The data report included 4 key charts. At the top there is a pie chart that shows the percentage of each species the data is corresponds to. 64% of the data is from loggerhead sea turtle measurements, 25.3% is from green sea turtles, and 10.4% is from ridley sea turtles. The percentage of data on hawksbill sea turtles was so small that it does not even show up on the chart, and to the left of the charts it can be seen that only 16 out of 5,698 sea turtles measured were hawksbill. Next, the left middle chart shows the number of sea turtles and species at different length measurements. For example, almost 3000 loggerhead sea turtles were measured to have a length close to 64.7 centimeters. The right middle chart is the same as the one before except it is for carapace width rather than length. Finally, the bottom chart shows the number of turtles at different weights for a sample of the data. It can be seen that green and ridley sea turtles generally weigh less than loggerhead sea turtles.

Finally, in part three, the process of training Cammy to identify the species of sea turtles began with uploading a Microsoft Excel CSV file that contained data with image URLs and the corresponding sea turtle species. Below is a small portion of this data from the uploaded file.

G1776144825_903aeeea86.jpg	[Green Sea Turtle]	G957175665_4e7c2b5e05.jpg	[Green Sea Turtle]
G179524988_1b49d05f69.jpg	[Green Sea Turtle]	Hdownload (11).jpeg	[Hawksbill Sea Turtle]
G2036633442_827c50a29d.jpg	[Green Sea Turtle]	Hdownload (13).jpeg	[Hawksbill Sea Turtle]
G2045183858_489f507954.jpg	[Green Sea Turtle]	Hdownload (7).jpeg	[Hawksbill Sea Turtle]
G2092972310_f23d2476fa.jpg	[Green Sea Turtle]	Himages (10).jpeg	[Hawksbill Sea Turtle]
G2231043515_c31f1eda01.jpg	[Green Sea Turtle]	Himages (12).jpeg	[Hawksbill Sea Turtle]
G2249269715_a16513d4b5.jpg	[Green Sea Turtle]	Himages (14).jpeg	[Hawksbill Sea Turtle]
G2351372108_1f392e7b34.jpg	[Green Sea Turtle]	Himages (2).jpeg	[Hawksbill Sea Turtle]
G2372187050_18be35a8d3.jpg	[Green Sea Turtle]	Himages (3).jpeg	[Hawksbill Sea Turtle]
G2383229369_ea14ac167d.jpg	[Green Sea Turtle]	Himages (4).jpeg	[Hawksbill Sea Turtle]
G2497314685_b0cb74d0c5.jpg	[Green Sea Turtle]	Himages (5).jpeg	[Hawksbill Sea Turtle]
G2504076508_3e6fe39ee6.jpg	[Green Sea Turtle]	Himages (7).jpeg	[Hawksbill Sea Turtle]
G2512587857_8eb7d68731.jpg	[Green Sea Turtle]	Himages.jpeg	[Hawksbill Sea Turtle]
G2518414896_a25d033622.jpg	[Green Sea Turtle]	Hmarine-species-green-turtle-istock-spd0218.jpeg	[Hawksbill Sea Turtle]
G2561708839_6db63bd9d9.jpg	[Green Sea Turtle]	L119755.jpeg	[Loggerhead Sea Turtle]
G2572114900_0d7f96b84a.jpg	[Green Sea Turtle]	L1343970215_afac0d49c2.jpg	[Loggerhead Sea Turtle]
G2655083390_31bfe0c85d.jpg	[Green Sea Turtle]	L178040052_fdbb3ba44b.jpg	[Loggerhead Sea Turtle]
G487076519_3eb1d25a8d.jpg	[Green Sea Turtle]	L193390782_0f03ccb174.jpg	[Loggerhead Sea Turtle]

Figure 4: Sample of Sea Turtle Image Data

This data file contained images of 60 different sea turtles. The images were labeled with the corresponding species in an external app and then downloaded again to create the table shown above. However, before training Cammy to use this data, it was important for me to better understand the data myself. Therefore, I made the data report shown below on Google Looker Studio.

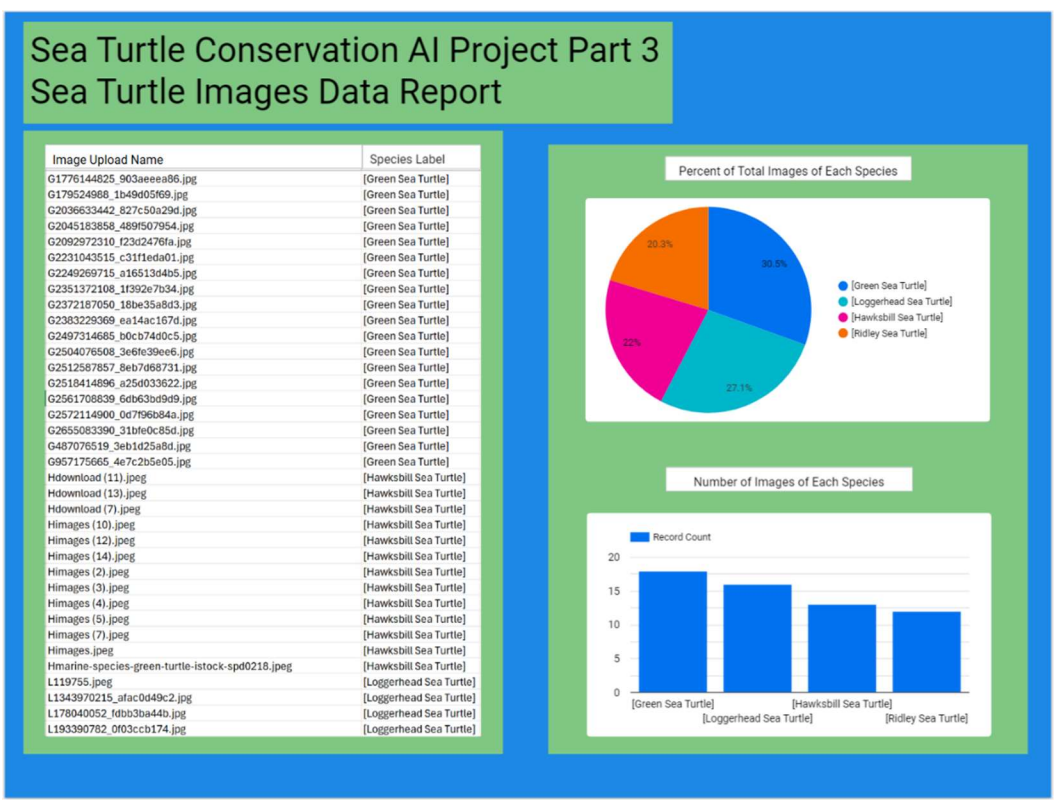

Figure 5: Sea Turtle Image Data Report

The data report included 2 key charts in addition to the table from figure 4. The pie chart in the top right shows the percentage of each species the data corresponds to. The chart shows that 27.1% of the images are of loggerhead sea turtles, 30.5% are of green sea turtles, 20.3% are of ridley sea turtles, and 22% are of hawksbill sea turtles. The bottom right chart shows the number of images of each species the data contains. There are roughly 18 green sea turtle images, 16 loggerhead sea turtle images, 13 hawksbill sea turtle images, and 12 ridley sea turtle images.

Model Conceptualization & Presentation

In the beginning of training for part one, Cammy had zero knowledge on sea turtles when asked about them as shown by "Answer 1" in Figure 6 below.

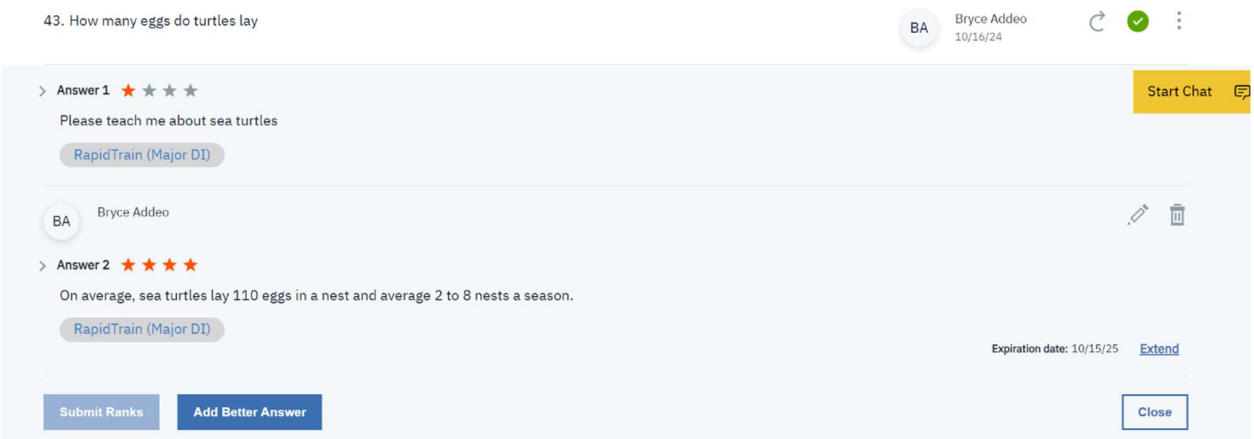

Figure 6: Answers to the first questions

The first question the bot was trained on was "How many eggs do turtles lay?". Since Cammy was completely untrained at this point, it initially responded with "Please teach me about sea turtles". To train the bot, its answers had to be given a ranking of one star to four stars, with one star being completely irrelevant and four stars being the absolute correct answer. Therefore, the first answer was given a rating of one star. To train Cammy to provide better answers, I had to find the right answers on google and then type them into the bot's database. This correct answer then appeared as "Answer 2" as shown above and was given a ranking of four stars. Therefore, Cammy now knew that this was the right answer to provide for a question asking about the number of eggs that sea turtles lay. However, as soon as the next different question was asked, Cammy gave this same answer because it was the only knowledge it had about sea turtles as shown below in Figure 7.

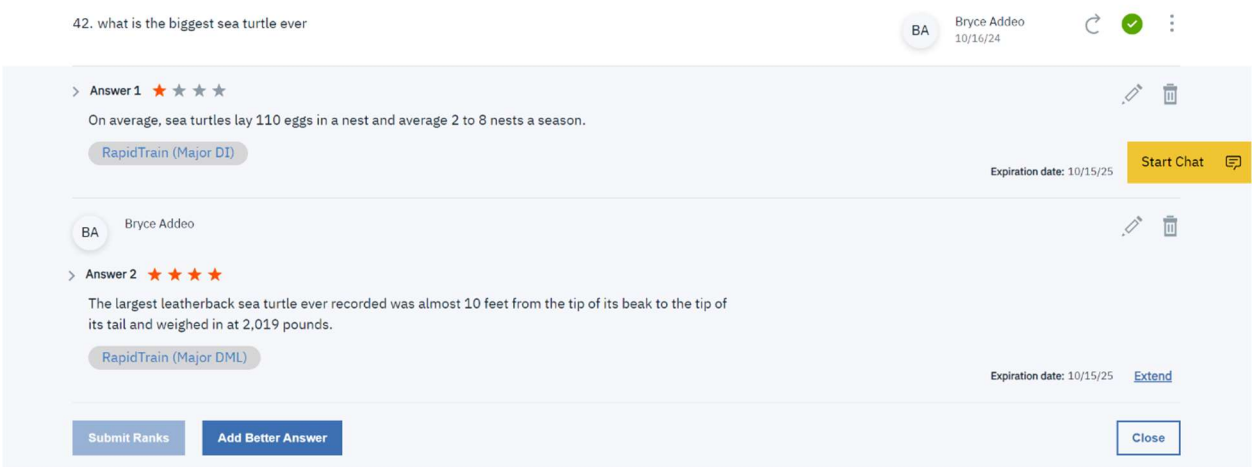

Figure 7: Answers to the second question

Obviously, the number of eggs is not relevant to a question asking about the biggest sea turtle, and therefore this answer that was given four stars for the first question was now given one star for the second question. In the same fashion as before, the correct answer was found on google and provided to the bot with a four star ranking to train it to answer questions about the size of the biggest sea turtle. This process was repeated for all unique questions. However, once similar questions began popping up, Cammy would provide the correct answer from its database on the first try since it would recognize similar key words from previous questions already asked. This was evidence that the training was working as the bot was learning to answer questions about sea turtles without assistance.

Next, in part two, the first step in training Cammy to predict weights of sea turtles was to run an experiment using the regression algorithm AutoKeras. Figures 8 and 9 below show the data being selected for the experiment as well as the dependent and independent variables being selected, respectively.

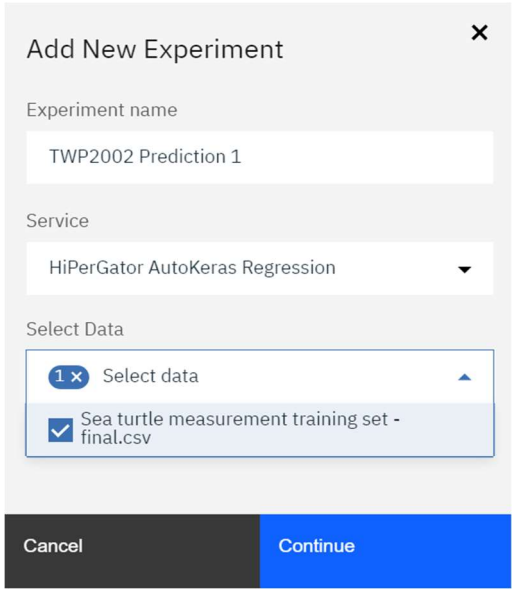

Figure 8: Data Upload to Experiment

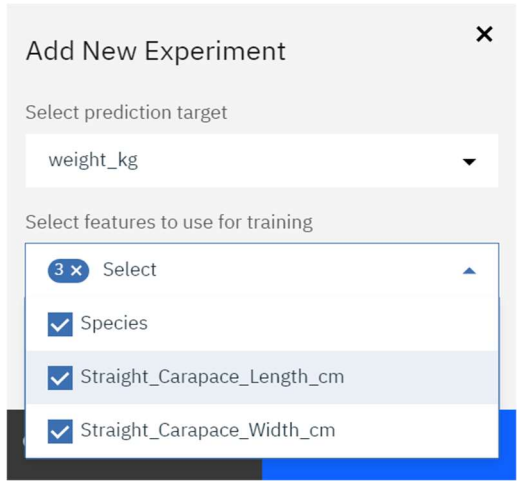

Figure 9: Variable Selection

Once these two steps were done, the algorithm was completed so that a relationship was created between the measurement variables to predict the weight of sea turtles. The last step was to upload this algorithm to Cammy which was done with the Model ID code as shown in the figure below.

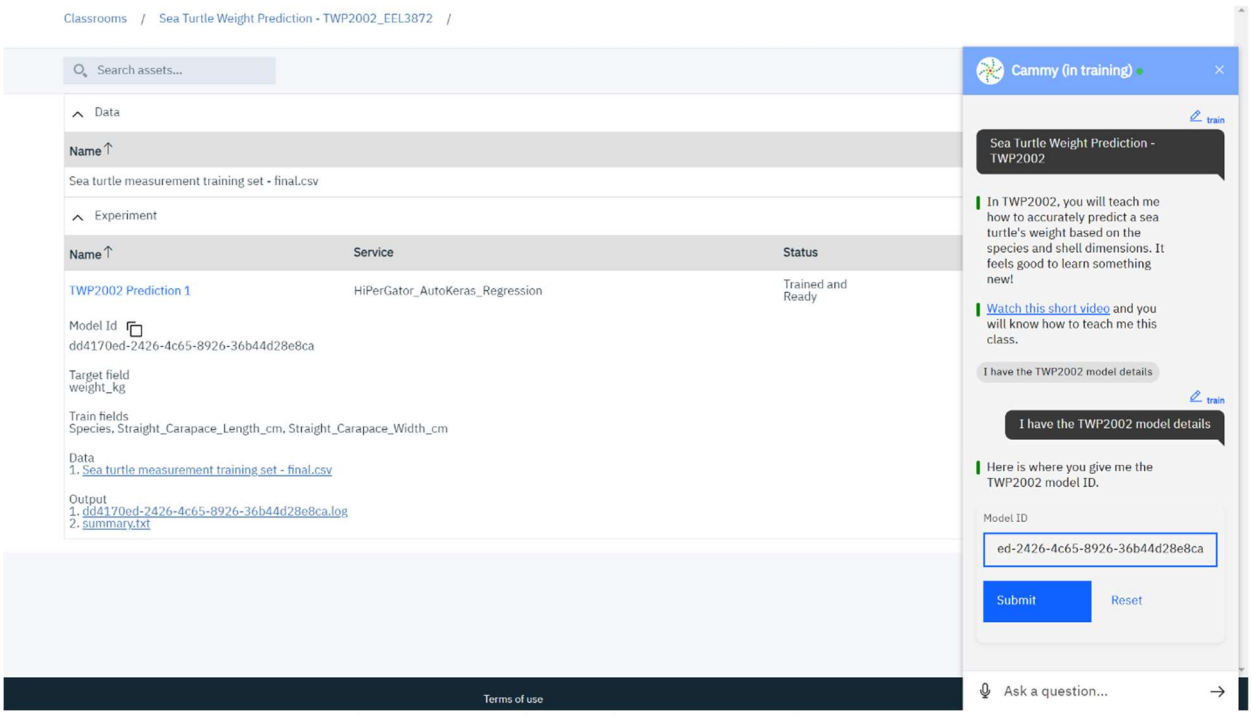

Figure 10: Uploading Algorithm to Cammy with Model ID

Finally, to train Cammy to label images of sea turtles in part three, I had to run an experiment using the algorithm AutoKeras. Figures 11 and 12 below show the data being uploaded and selected for the experiment, respectively.

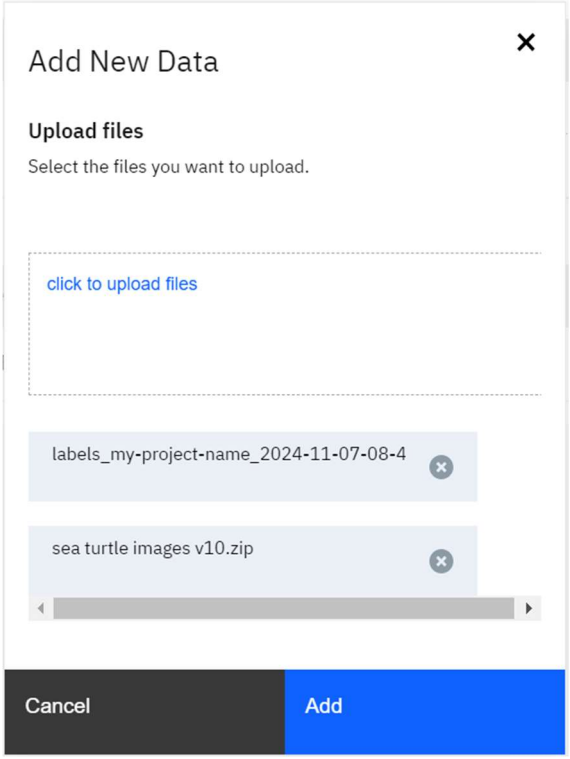

Figure 11: Uploading Data

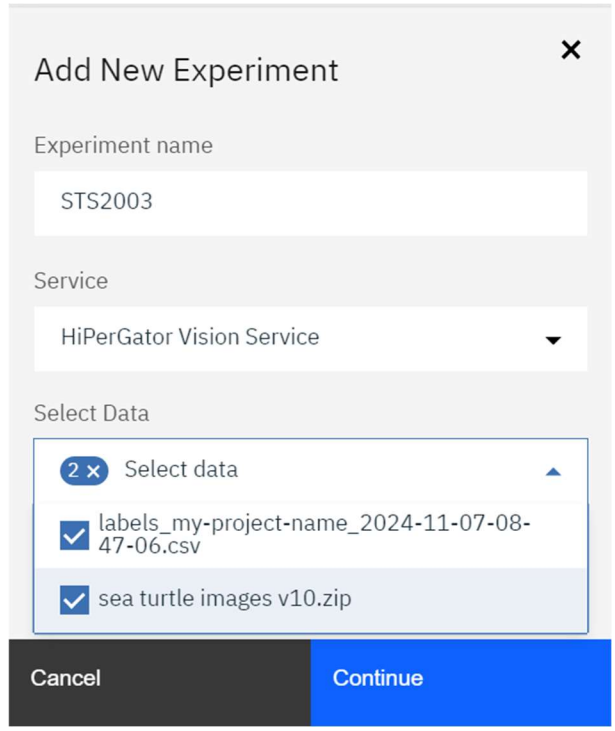

Figure 12: Selecting Data for Experiment

Once these two steps were done, the algorithm was created and ready to upload to Cammy by copying and pasting the Model ID code shown in the figure below.

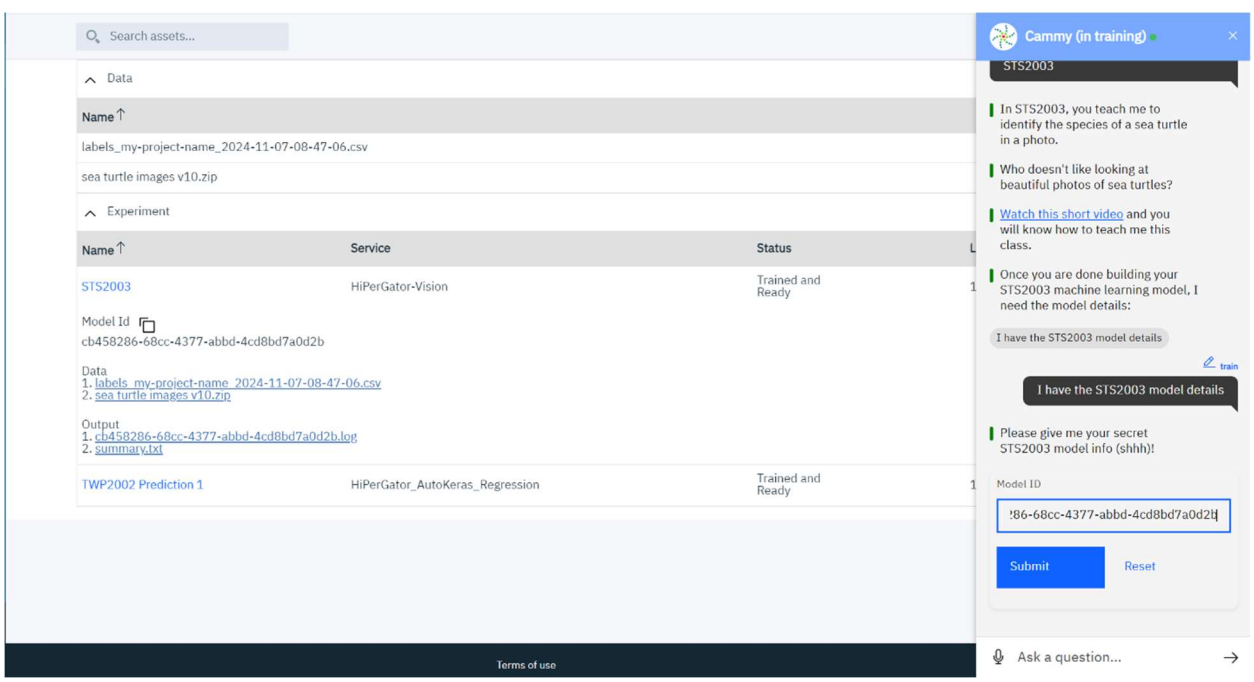

Figure 13: Uploading Algorithm to Cammy with Model ID

Verification and Validation

After the answers to all 40 questions were ranked in part one, I let Cammy rest and then tested it the next day with six of the questions it was trained on as shown in Figure 14 below.

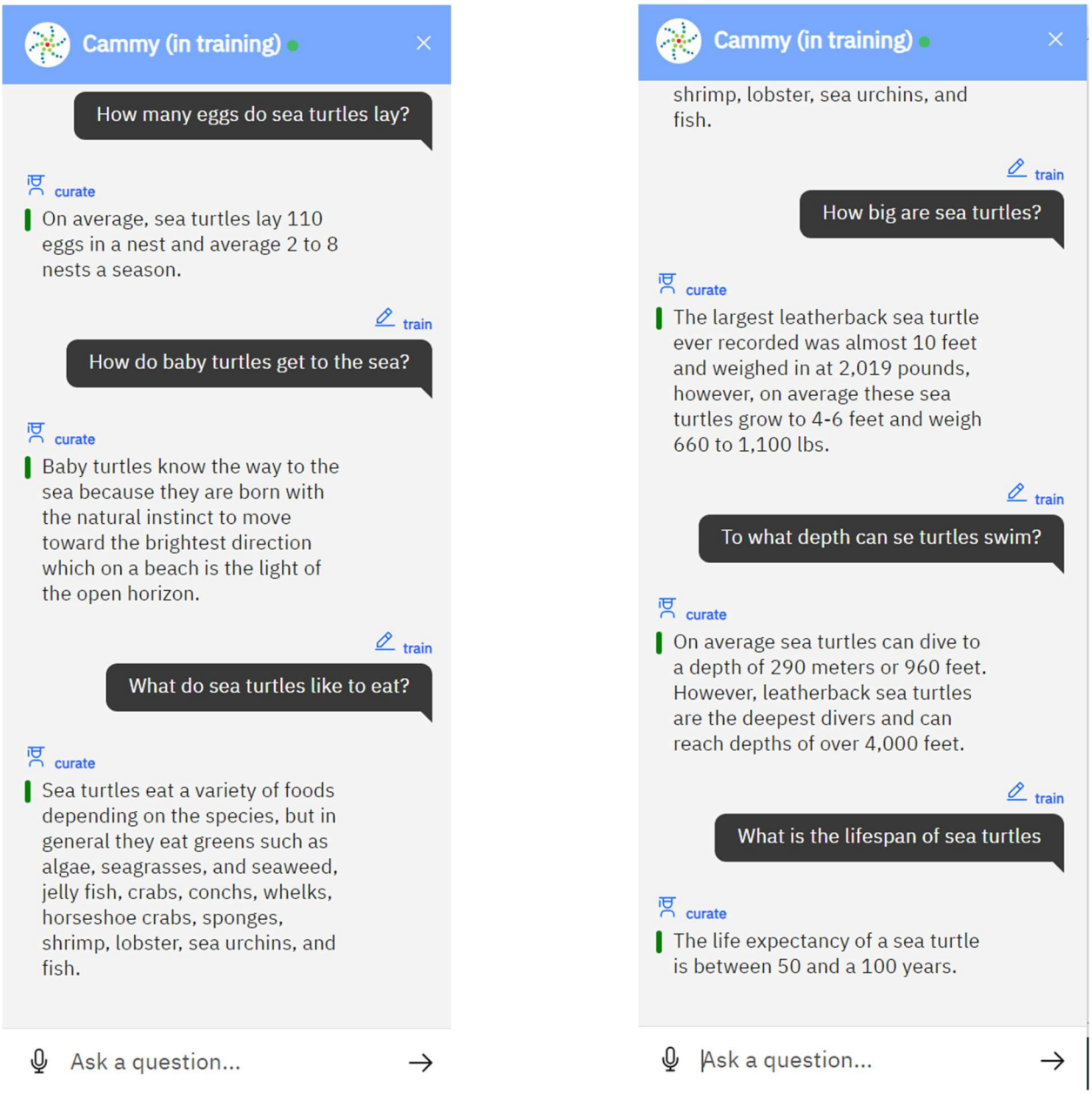

Figure 14: Testing Cammy with Six Questions (Part 1)

Even though the questions were worded slightly differently and some contained typos, Cammy was still able to answer them perfectly based on its training. This is important because most people will ask questions in different ways and with typing errors, so Cammy needs to be able to accurately answer questions regardless of these variables. It was able to do this by identifying key words that it has seen in previous questions and understanding the English language from its training.

For part two, once the algorithm was fully uploaded to Cammy, it was now time to test it out. I inputted the same length and width measurements for all four sea turtle species and Cammy calculated the weights as shown in the figure below.

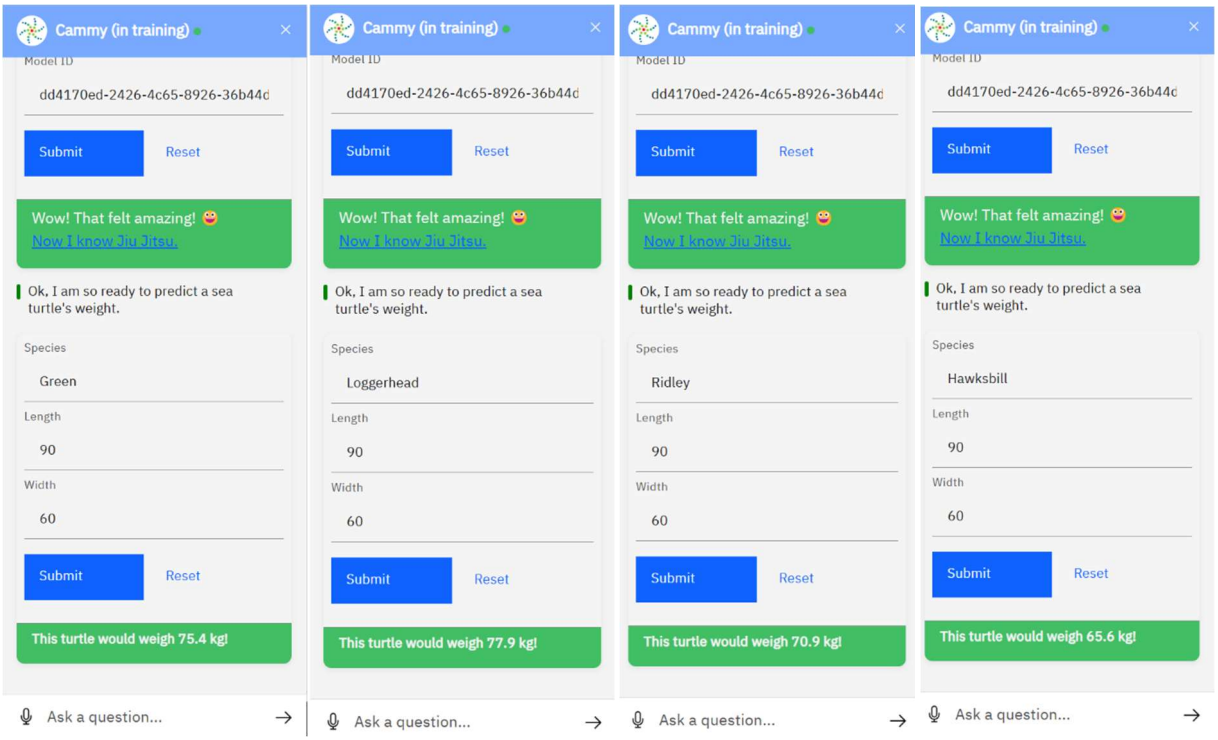

Figure 15: Testing Cammy (Part 2)

Despite the length and widths of the carapace of all the turtles being the same, Cammy calculated different weights for the four different species of sea turtles with the loggerhead being the heaviest at 77.9 kg and the hawksbill being the lightest at 65.6 kg. This makes sense because different sea turtle species have different sized features outside of their shells such as flippers. Additionally, the biological density of the sea turtles likely varies with species contributing to the different weights for the same carapace dimensions.

For part three, once the algorithm was fully uploaded to Cammy, it was now time to test it out. I inputted four different image URLs from the Sea Turtle TEST Images and Cammy identified the

species and stated the probability of complete accuracy. The results are shown in the figure below.

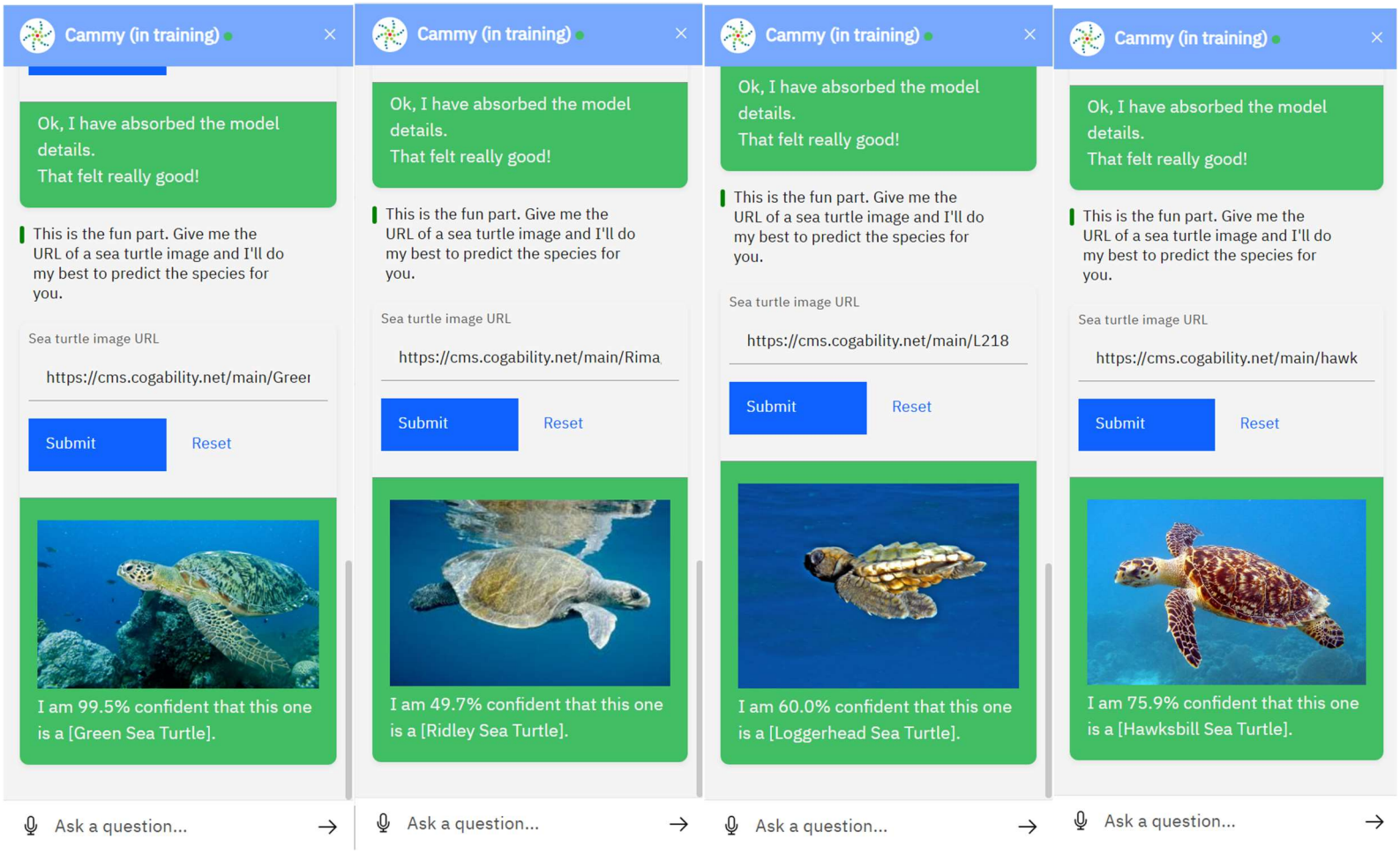

Figure 16: Testing Cammy (Part 3)

After receiving the input of the image URL, Cammy output the image along with a statement on identifying the species. From the examples shown, Cammy's confidence varies from 99.5% certainty to 49.7% certainty. This could likely be caused by differences in data between species as well as how similar certain species look to each other. While the above figures show all correct identifications, Cammy did not correctly identify the initial image of a loggerhead shown in the figure below.

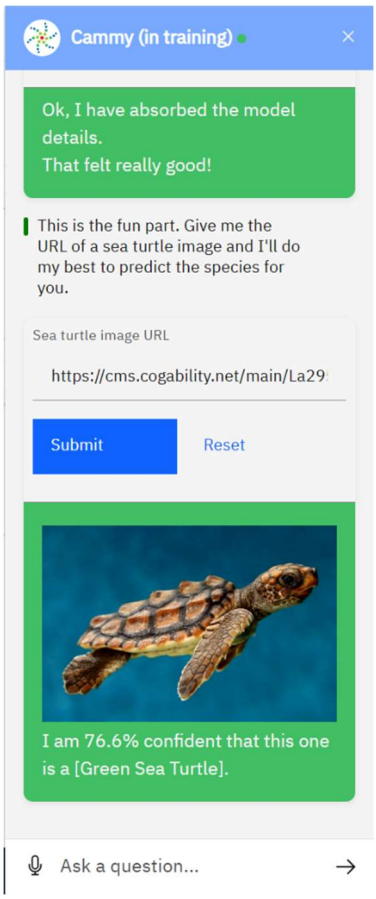

Figure 17: Incorrect Identification

Although this image showcases a young loggerhead sea turtle, Cammy identified it with 76.6% certainty as a green sea turtle. This seemed odd since all the other identifications were correct, however, this one was wrong despite Cammy's confidence being higher in this answer than some of the correct ones. Additionally, loggerhead sea turtles have unique features, especially compared to green sea turtles, which further contributes to the oddity of this mistake. Overall, this error proves that the algorithm and Cammy's training is not perfect and therefore the answers should not be taken with 100% certainty.

Experimentation

While Cammy was successfully trained to answer questions that most people would commonly ask about sea turtles, these questions did not consider some simpler things kids may be more curious about. Therefore, I added three new questions that kids may ask that Cammy did not know how to answer as shown in Figure 18 below. However, after training Cammy in the same

way as the first 40 questions, it quickly learned how to answer these questions too, shown in Figure 19. However, its punctuation and wording were slightly off, and this was likely because I asked again right after training instead of giving Cammy some time off to set the new answers in.

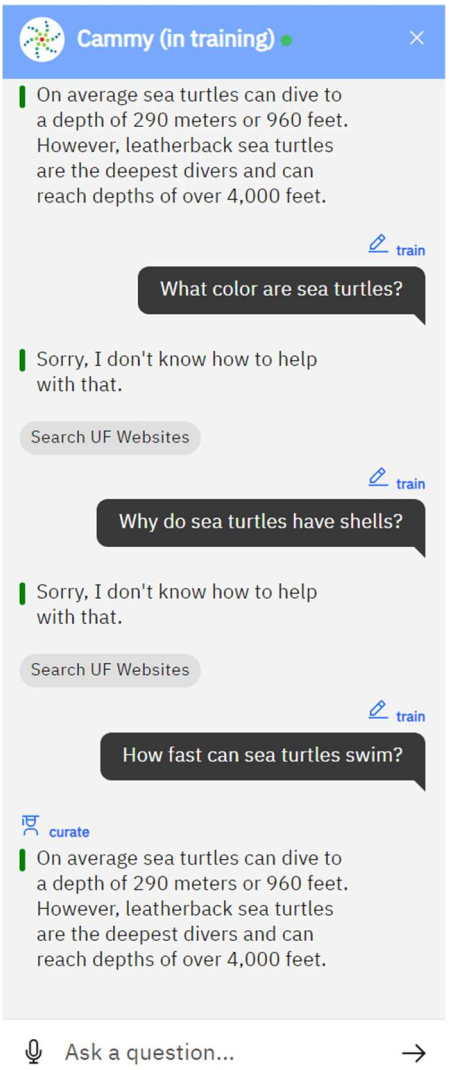

Figure 18: Kid Questions Before Training

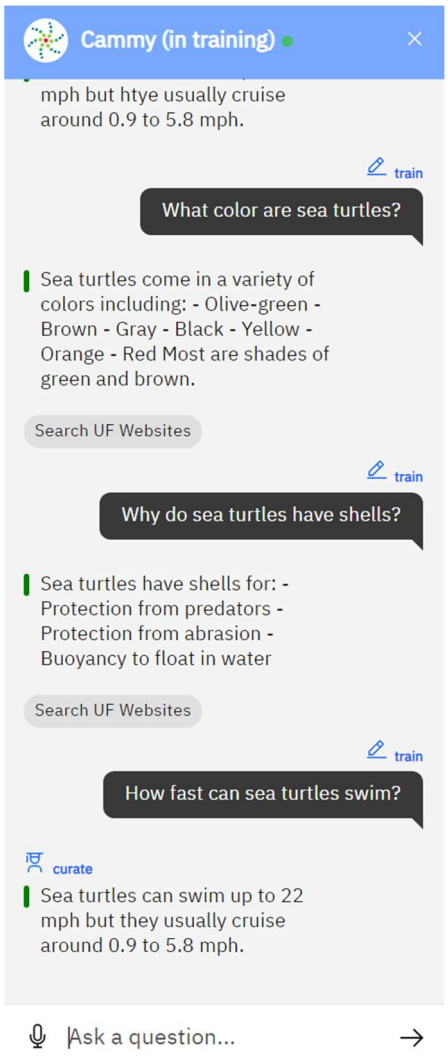

Figure 19: Kid Questions After Training

For part 3, an experiment was carried out to test Cammy's identification skills by uploading a random image URL from google and seeing if it could correctly identify the sea turtle species. Prior to this, Cammy identified the sea turtle species from specific test images within Cog University to almost perfect accuracy. The figures below show the image found on google and Cammy's output from the experiment.

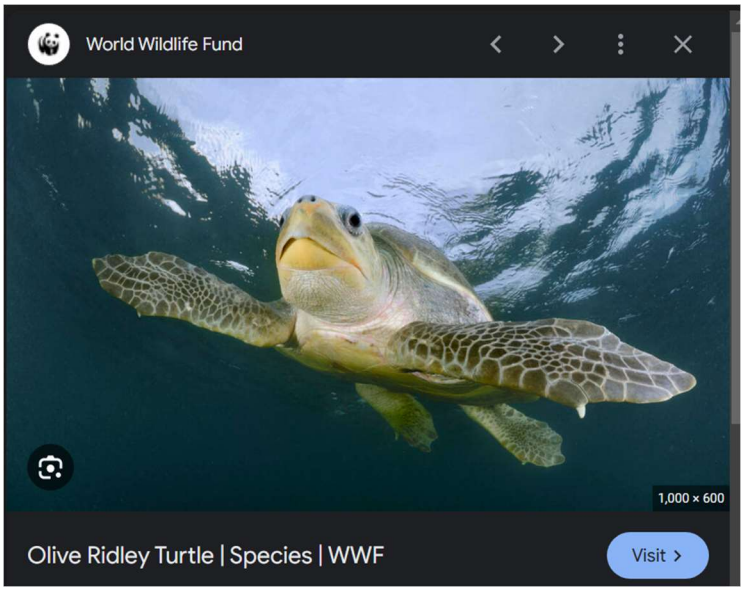

Figure 20: Random Google Sea Turtle Image

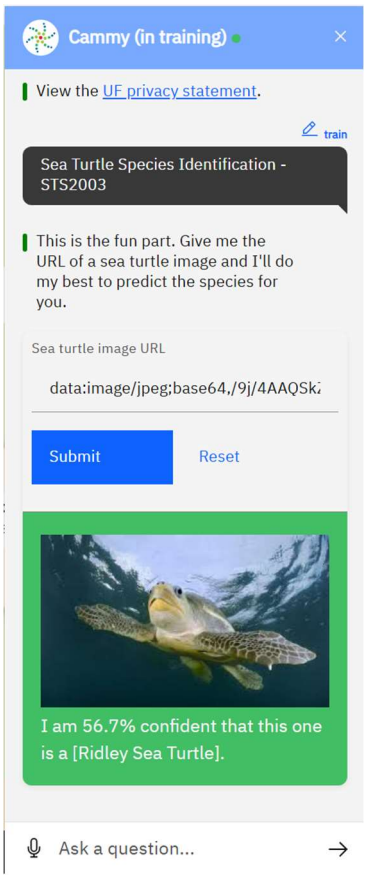

Figure 21: Testing Cammy on Random Image

As can be seen, Cammy correctly identified that the species of sea turtle in the random google photo was a ridley. This experiment proved that the test images were not just selected towards Cammy's identification strengths, but rather Cammy's training prepared it to identify any sea turtle image accurately.

Summary and Recommendations

In part one, the CogBot, Cammy, was trained to answer questions about sea turtles by learning from input responses. Once Cammy learned one kind of question it could recognize key words in a similar question with different wording or typing errors to provide the same correct answer without additional training. In the end, Cammy was able to perfectly answer over ten types of questions about sea turtles regardless of how they were worded or typing errors. Overall, this part of the project demonstrated how supervised learning can train an AI system to perform a certain task or achieve a certain goal.

Next, in part two, Cammy was trained to predict the weight of sea turtles when given their species, carapace length, and carapace width. This was possible due to a linear regression algorithm that analyzed data from nearly 6,000 sea turtles and created a relationship between the above variables. Once the algorithm was complete through UF's supercomputer, it was then uploaded to Cammy which was able to use it to predict sea turtle weights. The amount of data needed for machine learning regression analysis varies on the complexity of the model. In this case, it was a simple model but with multiple independent variables, so the amount of data was just right to complete the algorithm. Overall, the algorithm performed as intended and Cammy was now trained to output a weight prediction when given the species and carapace length and width of a sea turtle.

Finally, in part three, Cammy was trained to identify the species of sea turtle when given a sea turtle image URL. This was possible due to labeling image data on an external app and then uploading an algorithm that analyzed the data to Cammy. Once the algorithm was successfully uploaded, Cammy was able to identify any image of a sea turtle and give a percentage of how certain it was in the answer. Out of all images tested, Cammy only got one wrong, and this included random Google images in addition to the Cog University test images. Possible causes of this image recognition issue could be the amount of data that existed for loggerheads not being enough to fully train the model or similar features between loggerheads and green sea turtles that confused the algorithm. This problem could likely be resolved by uploading more data with sea turtles of various ages since the failure occurred on an image of a baby sea turtle. Overall, Cammy was successful in its training to identify sea turtle species based off images, however, further progress could still be made to increase precision and accuracy.

References

No external sources outside of Cog University were used for information in the content of this paper.

Appendix

All figures are included within the report.